Asthma and COPD

All key points for the exam (GINA, NAEPP, GOLD) + drug summary table

1. Asthma vs COPD — Overview

Feature	Asthma	COPD
Definition	**Chronic airway inflammation** with **variable, reversible** airflow obstruction and airway hyperresponsiveness	**Persistent** respiratory symptoms and **airflow limitation that is not fully reversible**, usually **progressive**
Onset	Often **childhood** (any age)	Usually **> 40 years**
Main cause / risk	Atopy/allergy, family history, triggers	**Smoking** (#1), biomass fuel, occupational dust, **alpha-1 antitrypsin deficiency**
Symptoms	**Variable** wheeze, dyspnea, chest tightness, cough — worse at **night/early morning**, with triggers	**Chronic, progressive dyspnea** (on exertion), chronic cough, **sputum production**
Spirometry	↓ FEV1/FVC with **significant bronchodilator reversibility** (↑ FEV1 > 12% **and** > 200 mL)	**Post-bronchodilator FEV1/FVC < 0.70** (persistent)
Inflammation	**Eosinophils**, mast cells, Th2 (IL-4, IL-5, IL-13), IgE	**Neutrophils**, macrophages, CD8 lymphocytes
Cornerstone therapy	**Inhaled corticosteroids (ICS)** — anti-inflammatory	**Long-acting bronchodilators** (LAMA, LABA)
LABA alone	**Never** (↑ asthma deaths) — always with ICS	Acceptable
ICS role	**Essential** for almost everyone	**Limited** (frequent exacerbations + eosinophils ≥ 100–300) — ↑ pneumonia risk

Asthma-COPD overlap: features of both → **always include an ICS** (treat like asthma) and **avoid LABA or LAMA monotherapy**.

2. Asthma: Diagnosis and Assessment

Triggers

Allergens (dust mites, pets, pollen, mold, cockroaches), **viral infections**, **tobacco smoke**, air pollution, **exercise**, cold air, strong odors, stress, **GERD**, **drugs** (see below), occupational exposures.

Drugs that can worsen asthma

- **Non-selective beta-blockers** (propranolol, nadolol, carvedilol, **timolol eye drops**) → bronchospasm • cardioselective (metoprolol, bisoprolol) are safer if needed.
- **Aspirin and NSAIDs → aspirin-exacerbated respiratory disease (AERD)**: asthma + nasal polyps + aspirin sensitivity.
- **ACE inhibitors** (cough), cholinergic drugs (bethanechol, pilocarpine), sulfites (preservatives).

Diagnosis

- **Variable symptoms** + **variable expiratory airflow limitation**.
- **Spirometry:** ↓ FEV1/FVC (< 0.75–0.80 in adults) + **reversibility**: FEV1 ↑ **> 12% and > 200 mL** after a bronchodilator.
- **Peak expiratory flow (PEF)** variability > 10% (adults) • bronchial challenge (**methacholine**) • **FeNO** (eosinophilic inflammation).
- Blood **eosinophils** and **IgE** help select **biologics**.

Assessing symptom control (GINA — past 4 weeks)

Question	Yes = 1 point
Daytime symptoms **more than twice a week**?	
Any **night waking** due to asthma?	
Reliever (SABA) needed **more than twice a week**?	(not counting pre-exercise use)
Any **activity limitation** due to asthma?	

0 = well controlled • 1–2 = partly controlled • 3–4 = uncontrolled • Asthma Control Test (**ACT**) score **≥ 20** = well controlled. **NAEPP "Rule of 2s":** symptoms or reliever use > 2 days/week, night waking > 2 times/month, or > 2 refills of SABA per year → not well controlled.

Severity classification (NAEPP, before treatment)

Severity	Symptoms	Night waking	SABA use	FEV1
Intermittent	≤ 2 days/week	≤ 2 times/month	≤ 2 days/week	> 80%
Mild persistent	> 2 days/week (not daily)	3–4 times/month	> 2 days/week (not daily)	> 80%
Moderate persistent	Daily	> once/week (not nightly)	Daily	60–80%
Severe persistent	Throughout the day	Often nightly	Several times/day	< 60%

3. Asthma: Stepwise Treatment (GINA 2024–2025, adults and adolescents ≥ 12)

Key change: SABA-only treatment is NOT recommended (↑ risk of severe exacerbations and death) — every patient should receive an **ICS-containing** treatment.

Step	Track 1 (PREFERRED): reliever = ICS-formoterol	Track 2 (alternative): reliever = SABA
Steps 1–2	**As-needed low-dose ICS-formoterol** (budesonide-formoterol) — "**AIR**" (anti-inflammatory reliever)	Step 1: take **ICS whenever SABA is taken** • Step 2: **daily low-dose ICS** + PRN SABA (or daily LTRA — less effective)
Step 3	**Low-dose ICS-formoterol maintenance AND reliever** (**MART**)	**Low-dose ICS-LABA** daily + PRN SABA
Step 4	**Medium-dose ICS-formoterol MART**	**Medium/high-dose ICS-LABA** + PRN SABA
Step 5	**Add-on LAMA** (tiotropium) • refer for phenotyping → **biologics** • high-dose ICS-formoterol	Add-on **LAMA** • **biologics** • (low-dose oral corticosteroids as last resort)

MART (maintenance and reliever therapy) — key points

- Only with **ICS-formoterol** (budesonide-formoterol or beclomethasone-formoterol) — formoterol has a **fast onset** like albuterol.
- **Budesonide-formoterol 160/4.5 mcg**: **1 inhalation as needed** (steps 1–2) • MART: **1–2 inhalations BID + 1 as needed**.
- **Maximum: 12 inhalations/day** (72 mcg formoterol) in adults.
- **Do NOT use MART** with other ICS-LABA (e.g., **fluticasone-salmeterol**) — salmeterol has a **slow onset**.
- **NAEPP 2020:** SMART (single maintenance and reliever) preferred for **steps 3–4** in ages **≥ 4**; intermittent ICS with SABA for mild persistent asthma (≥ 12); LAMA add-on (≥ 12) if not using SMART.

Other step principles

- **Before stepping up**: check **inhaler technique**, **adherence**, **trigger exposure**, comorbidities (rhinitis, GERD, obesity, smoking).

- **Step down** after **3 months** of good control (reduce ICS dose by 25–50%) — never stop ICS completely in adults.
- **Every patient** needs a **written asthma action plan**.
- **Albuterol-budesonide** (Airsupra) is an alternative ICS-SABA reliever (adults ≥ 18).

Children

- **6–11 years:** similar steps — daily low-dose ICS (step 2); low-dose ICS-formoterol MART or ICS-LABA (step 3–4); refer at step 5.
- **≤ 5 years:** PRN SABA • step 2: **daily low-dose ICS** (or intermittent short-course ICS at onset of viral illness) • use **MDI + spacer with face mask** (or nebulizer).
- **Montelukast** is an alternative in children but has a **boxed warning** for neuropsychiatric events.

ICS dose ranges (GINA, adults, total daily dose in mcg)

ICS	Low	Medium	High
Beclomethasone HFA (QVAR)	100–200	> 200–400	> 400
Budesonide DPI	200–400	> 400–800	> 800
Ciclesonide HFA	80–160	> 160–320	> 320
Fluticasone propionate (HFA/DPI)	100–250	> 250–500	> 500
Fluticasone furoate DPI	100	—	200
Mometasone DPI	200	400	≥ 800 (product-dependent)

Biologics (step 5, severe asthma)

Drug	Target	Use / dose
Omalizumab (Xolair)	**Anti-IgE**	**Allergic** asthma (IgE 30–700, positive skin/blood allergy test) • 75–375 mg SC every 2–4 weeks (dose by **IgE level and weight**) • **Boxed: anaphylaxis**
Mepolizumab (Nucala)	**Anti-IL-5**	**Eosinophilic** asthma (eos ≥ 150–300) • **100 mg SC every 4 weeks** • also EGPA, COPD (eosinophilic)
Reslizumab (Cinqair)	**Anti-IL-5**	Eosinophilic (eos ≥ 400) • **3 mg/kg IV every 4 weeks** • **Boxed: anaphylaxis**
Benralizumab (Fasenra)	**Anti-IL-5 receptor α**	Eosinophilic • **30 mg SC every 4 weeks × 3**, then **every 8 weeks**
Dupilumab (Dupixent)	**Anti-IL-4Rα** (blocks IL-4 and IL-13)	Eosinophilic or **OCS-dependent** asthma (also atopic dermatitis, nasal polyps, COPD) • 400 mg then 200 mg q2w, or 600 mg then 300 mg q2w • **Conjunctivitis**, transient **eosinophilia**
Tezepelumab (Tezspire)	**Anti-TSLP**	Severe asthma of **any phenotype** (including low eosinophils) • **210 mg SC every 4 weeks**

- Biologics are **add-on** therapy (continue inhalers) • check response after **4 months** • not for acute exacerbations • treat helminth infections first (anti-IL-5/IL-4 agents).

4. Asthma Exacerbations

Home management (action plan)

- **Peak flow zones: Green ≥ 80%** of personal best (continue) • **Yellow 50–79%** (caution: increase reliever, may need OCS) • **Red < 50%** (medical alert: SABA + seek emergency care).
- **SABA 2–4 puffs every 20 minutes for up to 1 hour** (or extra ICS-formoterol doses in MART patients, max 12/day).
- **Quadruple ICS dose** temporarily (GINA option) if on SABA reliever.
- Start **oral corticosteroids** if no improvement in 48 h or severe.

Emergency department / hospital

Treatment	Details
Oxygen	Target **SpO2 93–95%** (adults), 94–98% (children)
SABA	**Albuterol 4–10 puffs (MDI + spacer) every 20 minutes for 1 hour** (or 2.5–5 mg nebulized every 20 minutes × 3), then as needed
Ipratropium	**Add to SABA** for **moderate–severe** exacerbations in the **ED** (not after admission)
Systemic corticosteroids	**Prednisone 40–50 mg daily for 5–7 days** (children 1–2 mg/kg/day, max 40 mg, 3–5 days) • oral = IV • **no taper needed** if < 2 weeks • give within 1 hour
Magnesium sulfate IV	**2 g IV over 20 minutes** for **severe** exacerbations not responding to initial therapy
NOT recommended	**Antibiotics** (unless clear bacterial infection), **theophylline/aminophylline**, sedatives, routine chest X-ray
Discharge	**Start or step up ICS-containing controller** • check technique • action plan • **follow-up within 2–7 days**

Life-threatening signs: drowsiness, confusion, **silent chest**, cyanosis, exhaustion, rising PaCO2 → ICU, possible intubation.

5. Asthma: Special Situations

Situation	Key points
Exercise-induced bronchoconstriction	**SABA 5–20 minutes before exercise** or **ICS-formoterol PRN** before exercise • warm-up • LTRA (daily) is an alternative • frequent EIB = poor control → step up
Pregnancy	**Continue controllers** (uncontrolled asthma is more dangerous than the drugs) • **budesonide** = preferred ICS (most data) • albuterol = preferred SABA • montelukast acceptable
AERD (aspirin-exacerbated)	**Avoid aspirin and NSAIDs** (COX-2 inhibitors and acetaminophen < 1 g usually tolerated) • **LTRA** (montelukast) helpful • aspirin desensitization • dupilumab for nasal polyps
Elderly	Check technique and cognition • DPIs need a strong inspiratory flow • watch beta-agonist cardiac effects
Comorbidities	Treat **allergic rhinitis** (intranasal steroid), **GERD**, **obesity**, smoking, anxiety
Vaccines	**Influenza** yearly • **pneumococcal** • COVID-19 • RSV per age/risk

6. COPD: Diagnosis and Assessment (GOLD 2025)

Diagnosis

- **Symptoms** (dyspnea, chronic cough, sputum) + **risk exposure** (smoking ≥ 10 pack-years, biomass, occupational).
- **Post-bronchodilator FEV1/FVC < 0.70** confirms persistent airflow limitation (**required** for diagnosis).
- Screen **all COPD patients once** for **alpha-1 antitrypsin deficiency** (WHO) — especially young, non-smokers, lower-lobe emphysema.
- Subtypes: **chronic bronchitis** (productive cough ≥ 3 months for 2 consecutive years) and **emphysema** (alveolar destruction).

GOLD grades (airflow severity)

GOLD grade	Post-bronchodilator FEV1 (% predicted)
GOLD 1 (mild)	≥ 80%
GOLD 2 (moderate)	50–79%
GOLD 3 (severe)	30–49%
GOLD 4 (very severe)	< 30%

Symptom assessment

- **mMRC dyspnea scale** (0–4): **≥ 2** = more symptoms (stops for breath when walking at own pace on level ground).
- **CAT** (COPD Assessment Test, 0–40): **≥ 10** = more symptoms.

GOLD ABE groups (guide initial therapy)

Group	Exacerbation history (past year)	Symptoms	Initial therapy
A	0–1 moderate (no hospitalization)	**mMRC 0–1, CAT < 10**	**A bronchodilator** (long-acting preferred, except very occasional symptoms)
B	0–1 moderate (no hospitalization)	**mMRC ≥ 2, CAT ≥ 10**	**LABA + LAMA** (single inhaler preferred)
E	**≥ 2 moderate** or **≥ 1 leading to hospitalization**	Any	**LABA + LAMA** • consider **LABA + LAMA + ICS** if **blood eosinophils ≥ 300**

All patients: a **short-acting bronchodilator** for rescue • **LABA + ICS** is **not recommended** in COPD — if an ICS is indicated, use **triple therapy (LABA + LAMA + ICS)**.

7. COPD: Follow-up Pharmacologic Treatment

If dyspnea persists

- On a single long-acting bronchodilator → **LABA + LAMA**.
- Check **inhaler technique** and adherence • consider switching device or molecule • look for other causes (heart disease, anemia, deconditioning) • **pulmonary rehabilitation**.

If exacerbations persist

Current therapy	Next step
LABA or LAMA	**LABA + LAMA** (or LABA + LAMA + ICS if eos ≥ 300)
LABA + LAMA, **eos ≥ 100**	**Add ICS → triple therapy**
LABA + LAMA, **eos < 100**	**Roflumilast** (FEV1 < 50% + chronic bronchitis) or **azithromycin** (preferably former smokers)
Triple therapy	**Roflumilast** or **azithromycin** • **Dupilumab** (eos ≥ 300 + chronic bronchitis) • **Mepolizumab** (eos ≥ 300)
Patient on LABA + ICS	Escalate to **triple** if exacerbations (eos ≥ 100) • switch to LABA + LAMA if ICS not appropriate

When to use or withdraw ICS in COPD

Factor	Details
Strongly favors ICS	History of **hospitalization** for exacerbation • **≥ 2 moderate exacerbations/year** • **blood eos ≥ 300** • history of/concomitant **asthma**
Consider ICS	1 moderate exacerbation/year • eos 100–300
Against ICS	**Repeated pneumonia** • **eos < 100** • history of **mycobacterial infection**
Withdraw ICS	If **pneumonia**, inappropriate original indication, or no response (especially eos < 100)

Other COPD agents

- **Ensifentrine** (Ohtuvayre) — inhaled (nebulized) **PDE3/PDE4 inhibitor**, bronchodilator + anti-inflammatory, **3 mg BID** maintenance.
- **Theophylline** — not recommended routinely (narrow therapeutic index, many interactions).
- **Mucolytics** (N-acetylcysteine, carbocysteine) may ↓ exacerbations in some patients not on ICS.
- **Alpha-1 antitrypsin augmentation** (IV) for severe AATD with emphysema.
- **Opioids** (low dose) may relieve severe dyspnea in end-stage disease.

8. COPD: Non-Pharmacologic and Preventive Care

Intervention	Key points
Smoking cessation	**Most important intervention** — the only one shown to **slow lung function decline** and ↓ mortality
Pulmonary rehabilitation	Groups **B and E**, and after hospitalization • ↓ dyspnea, ↑ exercise capacity and quality of life
Vaccines	**Influenza** (yearly), **pneumococcal** (PCV20/PCV21), **COVID-19**, **RSV** (age ≥ 60, or younger with risk), **Tdap**, **zoster**
Long-term oxygen therapy (LTOT)	If **PaO2 ≤ 55 mmHg** or **SaO2 ≤ 88%** at rest (or PaO2 56–59 with cor pulmonale, heart failure, or polycythemia Hct > 55%) • use **≥ 15 hours/day** → **improves survival** • **no smoking** near oxygen (fire)
Non-invasive ventilation	Chronic hypercapnia; acute respiratory acidosis
Surgery	Lung volume reduction, bullectomy, **lung transplant**
Nutrition, activity	Maintain weight, regular physical activity

Smoking cessation pharmacotherapy

Drug	Dosing	Key points
Varenicline (most effective single agent)	Start **1 week before quit date**: 0.5 mg daily × 3 days → 0.5 mg BID × 4 days → **1 mg BID** × 12 weeks (may extend)	**Nausea** (take with food and water), **abnormal/vivid dreams**, insomnia • boxed warning for neuropsychiatric events was **removed (2016)** • renal adjustment
Nicotine replacement (NRT)	**Patch**: 21 mg/day if > 10 cigarettes/day (14 mg if ≤ 10) → taper over 8–12 weeks • **gum/lozenge**: 4 mg if first cigarette ≤ 30 min after waking, otherwise 2 mg	**Combination NRT** (patch + short-acting) is more effective • gum: "**chew and park**", no acidic drinks 15 min before • patch: skin irritation, vivid dreams (remove at night)
Bupropion SR	Start **1–2 weeks before quit date**: 150 mg daily × 3 days → **150 mg BID** × 7–12 weeks	**Contraindicated: seizure disorder**, **eating disorders**, abrupt alcohol/benzodiazepine withdrawal, **MAOIs** • insomnia, dry mouth • CYP2D6 inhibitor

Remember (CYP1A2): smoking (polycyclic hydrocarbons, not nicotine) **induces CYP1A2**. When a patient **quits smoking**, levels of **theophylline**, **clozapine**, **olanzapine**, and caffeine **rise** → monitor and reduce doses.

9. COPD Exacerbations (ECOPD)

- Acute worsening of **dyspnea**, **sputum volume**, and/or **sputum purulence** (within ≤ 14 days) • most often triggered by **viral or bacterial infection**, pollution.
- Severity: **mild** (SABD only) • **moderate** (SABD + oral steroids ± antibiotics) • **severe** (hospitalization/ED).

Treatment	Details
Short-acting bronchodilators	**SABA ± SAMA** (albuterol ± ipratropium) — MDI with spacer or nebulizer
Systemic corticosteroids	**Prednisone 40 mg daily for 5 days** • oral = IV • ↓ treatment failure, recovery time, relapse • blood eos can guide use
Antibiotics	If **increased sputum purulence** + increased dyspnea and/or sputum volume, or **mechanical ventilation** • **5–7 days** • choices: **amoxicillin-clavulanate**, **doxycycline**, **azithromycin** • risk of **Pseudomonas** (frequent antibiotics, severe disease) → **levofloxacin** or **ciprofloxacin**
Oxygen	Target **SpO2 88–92%** (avoid excess O2 → CO2 retention)
Non-invasive ventilation (NIV)	**First choice** for **acute respiratory acidosis** (pH ≤ 7.35 and PaCO2 ≥ 45 mmHg)
NOT recommended	**Theophylline**, routine mucolytics during exacerbation
Before discharge	Start/optimize **long-acting bronchodilator** maintenance • inhaler technique • pulmonary rehab • follow-up within 1–4 weeks

10. Drug Classes (Asthma and COPD)

Beta-2 agonists

Type	Drugs	Key points
SABA	**Albuterol** (salbutamol), **levalbuterol**	Rescue • onset 5 min, duration 4–6 h • frequent use (> 2 days/week; > 2 canisters/year) = poor control
LABA (12 h)	**Salmeterol**, **formoterol**, arformoterol	**Formoterol**: **fast onset** (can be a reliever in ICS-formoterol) • **Salmeterol**: **slow onset** (never for rescue)
Ultra-LABA (24 h)	**Indacaterol**, **olodaterol**, **vilanterol**	Once daily

- **ADRs: tremor**, **tachycardia**, palpitations, **hypokalemia**, **hyperglycemia**, nervousness, headache.
- **LABA monotherapy in asthma** → ↑ asthma-related deaths → **always combine with ICS** in asthma.
- **Non-selective beta-blockers** block the effect → avoid in asthma.

Anticholinergics (muscarinic antagonists)

Type	Drugs	Key points
SAMA	**Ipratropium**	QID • add-on in ED asthma exacerbation • COPD rescue (with albuterol)
LAMA	**Tiotropium**, **umeclidinium**, **aclidinium**, **glycopyrrolate**, revefenacin (nebulized)	**First-line maintenance in COPD** (LAMA is better than LABA at preventing exacerbations) • add-on step 5 in asthma (tiotropium)

- **ADRs: dry mouth** (most common), **urinary retention** (caution in **BPH**), constipation, **angle-closure glaucoma** (avoid spraying into eyes; use mouthpiece with nebulizer), metallic taste.
- **Do not combine** a LAMA with another LAMA, or ipratropium with a LAMA routinely.

Inhaled corticosteroids (ICS)

- **Drugs:** beclomethasone, **budesonide**, ciclesonide, **fluticasone** (propionate, furoate), mometasone.
- **Local ADRs: oral candidiasis (thrush)**, **dysphonia** (hoarseness) → **rinse mouth and spit** after each use, **use a spacer** with MDIs.

- **High-dose/long-term:** adrenal suppression, **osteoporosis**, **cataracts/glaucoma**, skin bruising, slight **growth reduction in children** (~1 cm, not progressive), **pneumonia in COPD** (especially fluticasone).
- **DDI: strong CYP3A4 inhibitors** (**ritonavir**, **cobicistat**, ketoconazole, itraconazole, clarithromycin) + **fluticasone, budesonide, mometasone** → **Cushing's syndrome** and adrenal suppression → use **beclomethasone** (or ciclesonide) instead.
- Full effect takes days to weeks → **use daily** (or as part of ICS-formoterol/ICS-SABA reliever) even when well.

Leukotriene modifiers

Drug	Key points
Montelukast (Singulair) — LTRA	Adults **10 mg once daily in the evening** (6–14 years 5 mg chewable; 2–5 years 4 mg) • **Boxed warning: serious neuropsychiatric events** (agitation, depression, **suicidal thoughts**, sleep disturbance) → reserve for patients who cannot use alternatives • useful in **AERD**, **allergic rhinitis**, exercise-induced bronchoconstriction
Zafirlukast (Accolate) — LTRA	20 mg BID on an **empty stomach** • **hepatotoxicity** • **CYP2C9 inhibitor** → ↑ **warfarin**
Zileuton (Zyflo) — 5-lipoxygenase inhibitor	600 mg QID (or CR 1,200 mg BID) • **hepatotoxicity** (monitor LFTs) • ↑ **theophylline**, ↑ **warfarin**, ↑ propranolol

Methylxanthines

- **Theophylline** — bronchodilator; **narrow therapeutic index**: target level **5–15 mcg/mL**.
- **Toxicity:** nausea, vomiting, insomnia, **tachyarrhythmias**, **seizures**.
- **CYP1A2 substrate**: ↑ levels with **ciprofloxacin**, **fluvoxamine**, **erythromycin/clarithromycin**, **cimetidine**, zileuton, **smoking cessation**, heart failure, liver disease • ↓ levels with **smoking**, **phenytoin**, **carbamazepine**, **rifampin**.

PDE-4 inhibitor (COPD only)

- **Roflumilast** (Daliresp): **250 mcg daily × 4 weeks** (better tolerance), then **500 mcg daily** • for **severe COPD (FEV1 < 50%) with chronic bronchitis and exacerbations** • **not a bronchodilator** (not for acute relief).
- **ADRs: diarrhea**, nausea, **weight loss** (monitor weight), headache, insomnia, **psychiatric effects/suicidality**.
- **Contraindicated:** moderate–severe liver impairment • **avoid strong CYP inducers** (rifampin, carbamazepine, phenytoin, phenobarbital) • CYP3A4/1A2 inhibitors ↑ levels • do not combine with theophylline.

Macrolide (COPD exacerbation prevention)

- **Azithromycin 250 mg daily** or **500 mg three times weekly** for 1 year in patients with frequent exacerbations (best in **former smokers**).
- **ADRs: QT prolongation** (baseline ECG), **hearing loss**, GI upset, antibiotic resistance.

11. Inhaler Devices and Counseling

Device	Technique (key steps)
Pressurized MDI (HFA)	**Prime** before first use (and if not used for a while) • **shake** • breathe out fully • seal lips, **press and inhale SLOWLY and deeply** (3–5 seconds) • **hold breath ~10 seconds** • wait **~1 minute** between puffs • **use a spacer** (ICS, children, elderly, poor coordination)
Dry powder inhaler (DPI)	**Do not shake** (most) • load dose • breathe out **away** from the device • inhale **QUICKLY and forcefully** (needs good inspiratory flow) • hold breath • **do not exhale into the device** • keep dry • **no spacer** • not ideal for young children or severe COPD with weak inspiration
Soft mist inhaler (Respimat)	Slow, deep inhalation (slow-moving mist, less coordination needed) • prime as directed

Device	Technique (key steps)
Nebulizer	For young children, elderly, severe exacerbations, poor technique • use mouthpiece with anticholinergics (eyes)

- **Rinse mouth and spit** after ICS.
- Use **rescue inhaler first**, then controller, if both are needed at the same time.
- Track doses (dose counter) • **check technique at every visit** (most common cause of poor control along with adherence).
- Bring the **written action plan** and know when to seek emergency care.

12. Common Combination Inhalers

Type	Product (generic)	Usual dose
ICS-LABA	**Symbicort** (budesonide/formoterol 80/4.5, 160/4.5)	2 inhalations BID • MART: 1–2 inh BID + 1 PRN (max 12/day)
	Advair (fluticasone propionate/salmeterol) Diskus 100/50, 250/50, 500/50	1 inhalation BID (COPD: 250/50) • **not for MART**
	Breo Ellipta (fluticasone furoate/vilanterol 100/25, 200/25)	1 inhalation once daily (COPD: 100/25)
	Dulera (mometasone/formoterol 100/5, 200/5)	2 inhalations BID
ICS-SABA	**Airsupra** (albuterol/budesonide 90/80)	2 inhalations PRN (max 12/day) — asthma reliever, adults
LAMA-LABA	**Anoro Ellipta** (umeclidinium/vilanterol 62.5/25)	1 inhalation once daily
	Stiolto Respimat (tiotropium/olodaterol 2.5/2.5)	2 inhalations once daily
	Bevespi (glycopyrrolate/formoterol 9/4.8)	2 inhalations BID
	Duaklir (aclidinium/formoterol 400/12)	1 inhalation BID
Triple (ICS-LAMA-LABA)	**Trelegy Ellipta** (fluticasone furoate/umeclidinium/vilanterol 100/62.5/25; 200 for asthma)	1 inhalation once daily
	Breztri (budesonide/glycopyrrolate/formoterol 160/9/4.8)	2 inhalations BID
SABA-SAMA	**Combivent Respimat** (ipratropium/albuterol 20/100)	1 inhalation QID (max 6/day)

Exam Traps and Frequently Tested Points

1. **Asthma** = variable, reversible (FEV1 ↑ > 12% and 200 mL) • **COPD** = post-bronchodilator **FEV1/FVC < 0.70**.
2. **SABA-only** treatment in asthma is **no longer recommended** — all patients need ICS.
3. **GINA preferred reliever: as-needed low-dose ICS-formoterol** (steps 1–2) and **MART** (steps 3–5).
4. **MART only with formoterol** (fast onset) — **never with salmeterol**; max **12 inhalations/day**.
5. **LABA monotherapy in asthma is contraindicated** (↑ deaths) • allowed in COPD.
6. **Uncontrolled:** > 2 days/week symptoms or reliever use, any night waking, activity limitation • **check technique and adherence before stepping up** • step down after 3 months of control.
7. **Asthma exacerbation:** albuterol every 20 min × 3, **ipratropium in ED (moderate–severe)**, **prednisone 40–50 mg × 5–7 days** (no taper), **IV magnesium 2 g** if severe • O2 93–95% • **no antibiotics/theophylline**.
8. **Peak flow:** green ≥ 80%, yellow 50–79%, red < 50%.
9. **EIB:** SABA (or ICS-formoterol) **5–20 minutes before exercise**.
10. **AERD:** avoid aspirin/NSAIDs; montelukast helps.
11. **Pregnancy:** continue therapy; **budesonide** preferred ICS.
12. **Biologics:** omalizumab (anti-IgE, allergic, **anaphylaxis**) • mepolizumab, reslizumab (IV, anaphylaxis), benralizumab (anti-IL-5/IL-5R, eosinophilic) • dupilumab (IL-4Rα; conjunctivitis) • **tezepelumab (TSLP; any phenotype)**.
13. **Montelukast:** boxed warning — **neuropsychiatric events** (suicidal thoughts).

14. **Zafirlukast/zileuton** → hepatotoxicity • ↑ warfarin (and zileuton ↑ theophylline).
15. **Theophylline 5–15 mcg/mL** • CYP1A2: **cipro, fluvoxamine**, macrolides, cimetidine, **quitting smoking** ↑ levels.
16. **COPD groups: A** → a bronchodilator • **B** → **LABA + LAMA** • **E** → **LABA + LAMA** (+ ICS if eos ≥ 300).
17. **ICS-LABA not recommended in COPD** → use triple if ICS indicated.
18. **ICS in COPD:** eos **≥ 300** strongly favors; **< 100** or **repeated pneumonia** → avoid/withdraw.
19. **Roflumilast:** FEV1 < 50% + chronic bronchitis; **diarrhea, weight loss, psychiatric effects**; avoid strong CYP inducers; not a bronchodilator.
20. **Azithromycin** long term → QT, hearing loss • best in former smokers.
21. **Dupilumab** and **mepolizumab** now approved for **eosinophilic COPD** • **ensifentrine** = nebulized PDE3/4 inhibitor.
22. **Smoking cessation** = most important COPD intervention • **varenicline** most effective • **bupropion** contraindicated in seizures/eating disorders.
23. **LTOT:** PaO2 ≤ 55 or SaO2 ≤ 88%, ≥ 15 h/day → ↑ survival.
24. **COPD exacerbation:** SABA ± SAMA, **prednisone 40 mg × 5 days**, antibiotics if purulent sputum (5–7 days), **O2 88–92%**, **NIV** for respiratory acidosis.
25. **Vaccines:** influenza, pneumococcal, COVID-19, **RSV (≥ 60)**, Tdap, zoster.
26. **ICS ADRs:** thrush, dysphonia → **rinse and spit**, use spacer • COPD → ↑ **pneumonia**.
27. **Ritonavir/cobicistat + fluticasone/budesonide** → Cushing's → use **beclomethasone**.
28. **Beta-agonist ADRs:** tremor, tachycardia, **hypokalemia**, hyperglycemia.
29. **Anticholinergic ADRs:** dry mouth, urinary retention (BPH), glaucoma.
30. **Non-selective beta-blockers** (including timolol eye drops) → avoid in asthma; cardioselective OK in COPD.
31. **MDI:** shake, inhale **slowly**, hold 10 s, spacer • **DPI:** no shake, inhale **fast and forceful**, no spacer, don't exhale into device.
32. **Asthma-COPD overlap** → always include ICS.

Note: GINA, NAEPP, and GOLD are updated often, and dose ranges vary by product. **Use the numbers in the reference you are studying for the exam.**

Drug Summary Table: Doses, Key ADRs and DDIs

Drug	Dose	Key ADRs	Key DDIs / special notes
Albuterol (SABA)	MDI 90 mcg/puff: 2 puffs q4–6h PRN • neb 2.5 mg • exacerbation: 4–10 puffs q20 min × 1 h	**Tremor**, tachycardia, **hypokalemia**, hyperglycemia	Blocked by **non-selective beta-blockers** • ↑ hypokalemia with diuretics, steroids
Levalbuterol (SABA)	MDI 45 mcg: 2 puffs q4–6h • neb 0.63–1.25 mg	Same (similar to albuterol)	R-isomer of albuterol
Ipratropium (SAMA)	MDI 17 mcg: 2 puffs QID • neb 0.5 mg (q20 min × 3 in ED)	Dry mouth, urinary retention, glaucoma (eye contact)	Anticholinergic burden • avoid routine use with LAMA
Salmeterol (LABA)	50 mcg DPI 1 inhalation BID	Tremor, tachycardia, headache	**Never alone in asthma** • **not for rescue** • avoid **strong CYP3A4 inhibitors** (QT, cardiac effects)
Formoterol (LABA)	12 mcg BID (DPI) • neb 20 mcg BID	Same	**Fast onset** → reliever in ICS-formoterol
Arformoterol / indacaterol / olodaterol / vilanterol	Arformoterol 15 mcg neb BID • indacaterol 75 mcg daily • olodaterol 5 mcg daily	Same as LABA	COPD (indacaterol, olodaterol, arformoterol) • vilanterol in combinations
Tiotropium (LAMA)	HandiHaler 18 mcg daily • Respimat 2 × 2.5 mcg (COPD) / 2 × 1.25 mcg (asthma) daily	**Dry mouth**, urinary retention, constipation, glaucoma	Do not swallow HandiHaler capsules • caution BPH, narrow-angle glaucoma

Drug	Dose	Key ADRs	Key DDIs / special notes
Umeclidinium / aclidinium / glycopyrrolate / revefenacin (LAMA)	Umeclidinium 62.5 mcg daily • aclidinium 400 mcg BID • revefenacin 175 mcg neb daily	Same as LAMA	COPD
Beclomethasone (ICS)	QVAR 40–320 mcg BID	**Thrush**, **dysphonia**; high dose: adrenal suppression, osteoporosis	Preferred ICS with **ritonavir/cobicistat** (less CYP3A4 effect) • breath-actuated (no spacer)
Budesonide (ICS)	DPI 180–360 mcg BID • neb 0.25–1 mg daily–BID (children)	Same	**Preferred in pregnancy** • **CYP3A4** (ritonavir → Cushing's)
Fluticasone propionate / furoate (ICS)	FP 88–880 mcg BID (HFA) • FF 100–200 mcg daily	Same + **pneumonia in COPD**	**Strong CYP3A4 inhibitors** (ritonavir, cobicistat, ketoconazole) → **Cushing's** — avoid
Mometasone / ciclesonide (ICS)	Mometasone 110–440 mcg daily–BID • ciclesonide 80–320 mcg BID	Same	Mometasone: CYP3A4 • ciclesonide: prodrug, less oral thrush
Budesonide/formoterol (Symbicort)	160/4.5: 1 inh PRN (steps 1–2) • MART 1–2 inh BID + PRN (max 12/day) • COPD 160/4.5 2 inh BID	ICS + LABA effects	**MART-capable** • rinse mouth
Fluticasone/salmeterol (Advair)	1 inh BID (100/50 to 500/50; COPD 250/50)	Same + pneumonia (COPD)	**Not for MART** • CYP3A4 inhibitors
Fluticasone furoate/vilanterol (Breo)	1 inh daily (100/25; 200/25 asthma)	Same	CYP3A4 inhibitors
Albuterol/budesonide (Airsupra)	2 inh PRN (max 12/day)	Headache, oral thrush, tremor	Asthma reliever (adults)
LAMA/LABA (Anoro, Stiolto, Bevespi, Duaklir)	Anoro 1 inh daily • Stiolto 2 inh daily • Bevespi 2 inh BID • Duaklir 1 inh BID	Anticholinergic + beta-agonist effects	COPD first-line for groups B and E
Triple (Trelegy, Breztri)	Trelegy 1 inh daily • Breztri 2 inh BID	Thrush, pneumonia, dry mouth	COPD with exacerbations + eos ≥ 100 • Trelegy 200 also for asthma
Montelukast	10 mg daily in evening (adults) • 5 mg (6–14 y) • 4 mg (2–5 y)	**Boxed: neuropsychiatric events** (suicidality, sleep disturbance)	↓ levels with phenobarbital, rifampin
Zafirlukast	20 mg BID, empty stomach	**Hepatotoxicity**, headache	**CYP2C9 inhibitor** → ↑ **warfarin** • food ↓ absorption
Zileuton	600 mg QID (or CR 1,200 mg BID)	**Hepatotoxicity**, neuropsychiatric	↑ **theophylline**, ↑ **warfarin**, ↑ propranolol
Theophylline	300 mg/day → titrate to level **5–15 mcg/mL**	Nausea, insomnia, **arrhythmias**, **seizures**	**CYP1A2**: ↑ cipro, fluvoxamine, macrolides, cimetidine, **smoking cessation** • ↓ smoking, phenytoin, carbamazepine, rifampin
Roflumilast	250 mcg daily × 4 wk → 500 mcg daily	**Diarrhea**, **weight loss**, nausea, **psychiatric/suicidality**	**Avoid strong CYP inducers** • CI moderate–severe hepatic impairment • not with theophylline
Azithromycin (COPD prevention)	250 mg daily or 500 mg 3 × weekly	**QT prolongation**, **hearing loss**, GI	Avoid with other QT drugs
Ensifentrine (Ohtuvayre)	3 mg nebulized BID	Back pain, hypertension, diarrhea, psychiatric events	COPD maintenance

Drug	Dose	Key ADRs	Key DDIs / special notes
Omalizumab	75–375 mg SC q2–4 wk (by IgE and weight)	**Boxed: anaphylaxis**, injection site reactions	Observe after doses; **epinephrine autoinjector**
Mepolizumab	100 mg SC q4 wk	Headache, injection site reactions, herpes zoster	Treat helminth infections first • asthma, COPD, EGPA
Reslizumab	3 mg/kg IV q4 wk	**Boxed: anaphylaxis**	IV only
Benralizumab	30 mg SC q4 wk × 3, then q8 wk	Headache, pharyngitis	Eosinophil depletion
Dupilumab	Asthma: 400 mg then 200 mg q2w (or 600/300) • COPD: 300 mg q2w	**Conjunctivitis**, **eosinophilia**, injection reactions	Avoid live vaccines
Tezepelumab	210 mg SC q4 wk	Pharyngitis, arthralgia, back pain	Any phenotype
Prednisone	Asthma exacerbation 40–50 mg × 5–7 days • COPD exacerbation 40 mg × 5 days	Hyperglycemia, mood changes, insomnia, fluid retention	No taper if < 2 weeks • ↑ GI bleed with NSAIDs
Methylprednisolone IV	40–80 mg/day (if unable to take oral)	Same	Oral is as effective as IV
Magnesium sulfate IV	2 g IV over 20 min (severe asthma exacerbation)	Flushing, hypotension	Caution in renal impairment
Varenicline	0.5 mg daily × 3 d → 0.5 mg BID × 4 d → 1 mg BID × 12 wk	**Nausea**, **vivid dreams**, insomnia	Renal adjustment (CrCl < 30) • start 1 week before quit date
Bupropion SR	150 mg daily × 3 d → 150 mg BID × 7–12 wk	Insomnia, dry mouth, **seizures**	**CI: seizures, eating disorders, MAOIs • CYP2D6 inhibitor** (↑ metoprolol, TCAs; ↓ tamoxifen activation)
Nicotine patch / gum / lozenge	Patch 21 mg (> 10 cig/day) or 14 mg, taper • gum/lozenge 2–4 mg PRN	Skin irritation, vivid dreams (patch) • jaw pain, hiccups (gum)	Gum: "chew and park", no acidic drinks 15 min before • combination NRT more effective

Good luck on your exam!